



HOUSTON, TEXAS

Predicting tomorrow's **air quality.**

A machine learning project that forecasts the next-day Air Quality Index category for Houston using historical pollution and weather data.

End-to-end pipeline · Multi-class classification

MSAIS 5302 - AI & Machine Learning
Final Presentation
Mitaali Dayal

Why predict tomorrow's air?

Today's AQI is already public. Tomorrow's hasn't happened yet — and that's the number people actually need to make decisions.



Parents

Should my asthmatic child play outside tomorrow?



Schools

Do we hold tomorrow's outdoor PE class?



City teams

Should we issue a public-health advisory?



Athletes

Is tomorrow's morning run safe to do outside?

Same forecast, four different decisions. The model exists to help all of them.

Houston's air quality keeps making headlines.

Recent reporting on the same problem we're trying to forecast — from rankings, ozone alerts, refinery emissions, and wildfire smoke.

AMERICAN LUNG ASSOCIATION ·
STATE OF THE AIR 2025

**Houston metro ranked 7th most
polluted U.S. city for ozone.**

NEWSWEEK · 2024

**Houston saw nearly 50 ozone
action days in 2024.**

HOUSTON PUBLIC MEDIA · AUG 2024

**"Dirty Dozen" report names city's
12 worst air polluters.**

HOUSTON LANDING · MAY 2024

**Wildfire smoke from Central
America blankets Houston.**

TEXAS TRIBUNE · MAR 2024

**Air monitors fail Latino
communities near the Ship
Channel.**



WHY THIS PROJECT EXISTS

Houston has a real, recurring air quality problem. A reliable next-day forecast helps families and schools plan ahead.

Three categories, one prediction.

The Air Quality Index runs 0–500. We group the relevant range into three actionable classes.

0 – 50



Good

Air is clean. Go outside, exercise, breathe deep.

51 – 100



Moderate

Acceptable for most. Some sensitive folks may notice it.

101+



Unhealthy for sensitive groups

Asthmatics, kids, elderly, people with heart conditions should reduce outdoor exertion.

We frame this as multi-class classification: given today, predict which class tomorrow falls into.

Two public sources, six years.

Both sources pulled directly from public APIs — no manual downloads, fully reproducible.



POLLUTION

EPA Air Quality System

Daily AQI values for Harris County, TX (Houston). Includes the dominant pollutant — PM2.5, ozone, PM10, NO₂.



WEATHER

Open-Meteo Archive

Daily temperature, humidity, wind speed, and precipitation at Houston's coordinates.

Joined on date

6 years

of daily history

2,192 days

of observations

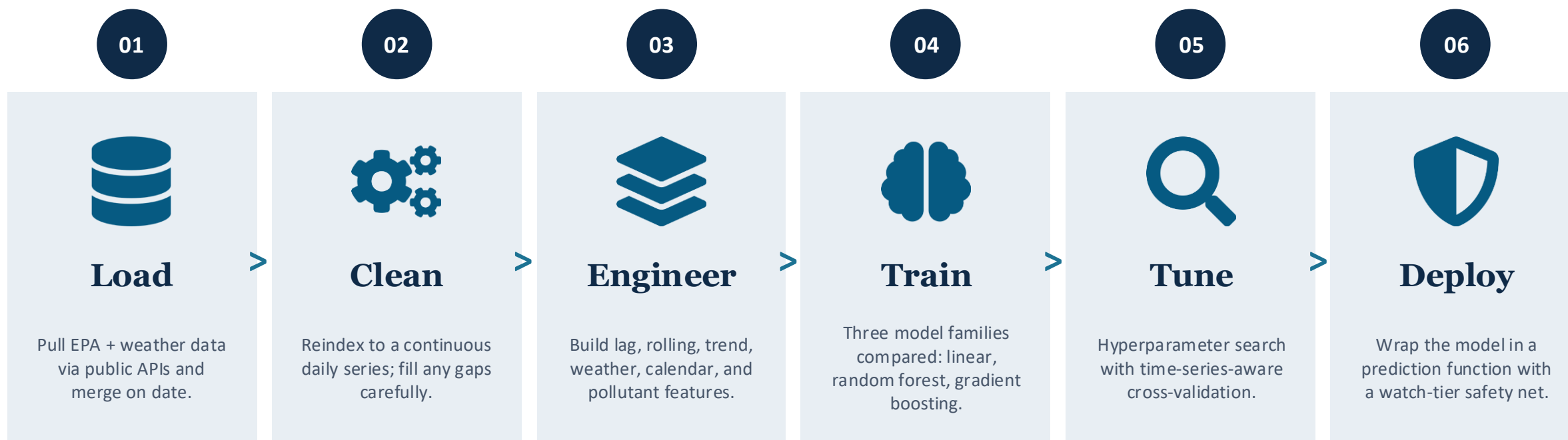
Houston

Harris County, TX

Two domains

pollution + weather

How we got from raw data to a forecast.



Each step was validated honestly: time-based splits, no leakage, diagnostics before declaring success.

Transforming raw data into a usable format

Sorted all records by date.

Checked for missing daily records.

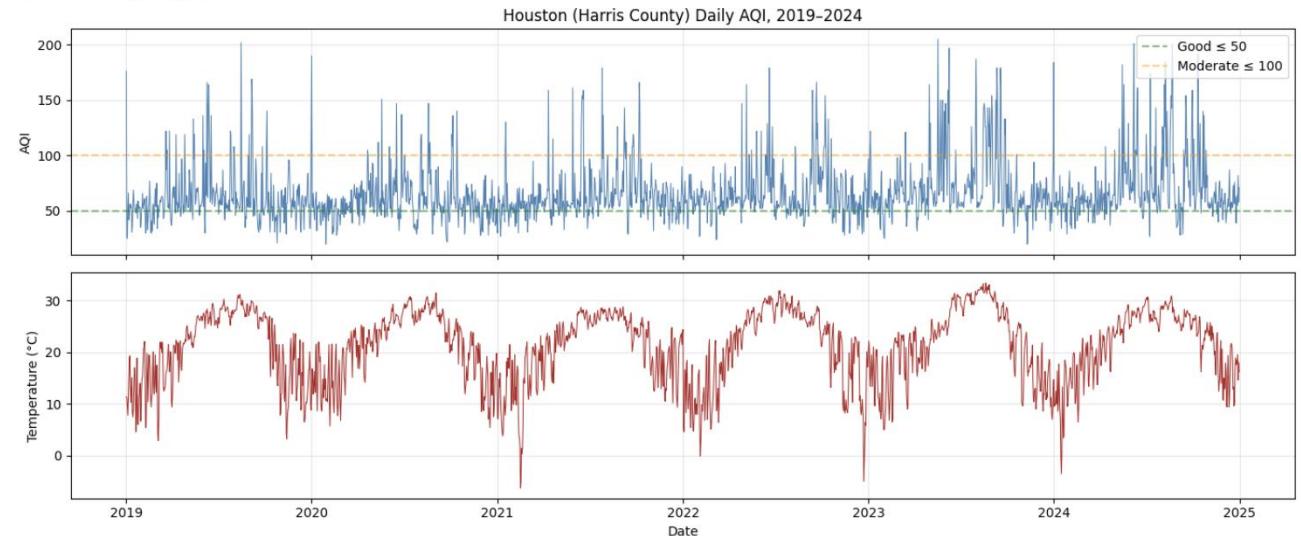
Created a continuous daily dataset.

Filled weather gaps using forward/backward fill. Interpolated AQI values if gaps existed.

📅 Date range: 2019-01-01 → 2024-12-31
Expected days: 2192 | Actual: 2192 | Missing: 0
✅ Preprocessing complete: 2192 continuous daily rows, no missing values

📊 AQI category distribution (peek at class balance):
AQI
0_Good 18.9 %
1_Moderate 72.4 %
2_USG+ 8.7 %
Name: proportion, dtype: object

📊 Defining pollutant distribution:
defining_parameter
PM2.5 1513
Ozone 568
PM10 85
NO2 26
Name: count, dtype: int64



AQI changes over time, while temperature shows seasonal patterns. Most days are Moderate, but harmful USG+ days still appear as spikes.

Three model families, three different bets.

We compared models with very different assumptions to see which best fits this problem.

B A S E L I N E

Logistic Regression

- *Linear decision boundaries.*

A simple, interpretable starting point. Fast to train and easy to reason about — perfect for setting a floor.

E N S E M B L E

Random Forest

- *Many independent decision trees.*

Captures nonlinear interactions automatically. Robust to noise and outliers, hard to overfit catastrophically.

B O O S T I N G

LightGBM

- *Trees that learn from each other's mistakes.*

Sequentially fits trees to fix prior errors. Usually the strongest performer on structured tabular data.

Each one tells us something different — together they bracket what's actually possible on this problem.

What does the model see?

Goal is to capture patterns affecting air quality



History

- Yesterday, 2-day, and 7-day AQI
- 3- and 7-day rolling averages
- Volatility over the past week



Trend

- Day-over-day AQI change
- Multi-day momentum signals
- Is air quality rising or falling?



Weather

- Temperature, humidity, wind, rain
- Yesterday's conditions as lags
- Hot + still air interaction term



Calendar

- Day of week, weekend flag
- Cyclical month encoding
- Day-of-year for seasonality

Model Performance

We compared models with very different assumptions to see which best fits this problem.

Winner

Random Forest

Random Forest performed best

Accuracy $\approx$ 72–74%

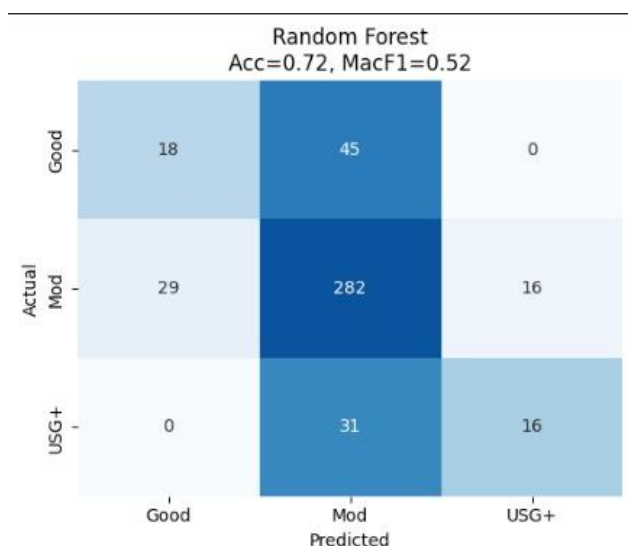
Macro F1 $\approx$ 0.52

Key Insights:

Model predicts “Moderate” very well

Struggles with USG+ (harmful days)

Accuracy alone is misleading due to class imbalance



Model Improvements:

- Hyperparameter tuning improved Macro F1 to $\sim$ 0.57
- Better detection of USG+ (harmful days)
- Slight drop in overall accuracy
- Used regularized RF + threshold tuning
- Balanced accuracy and risk detection

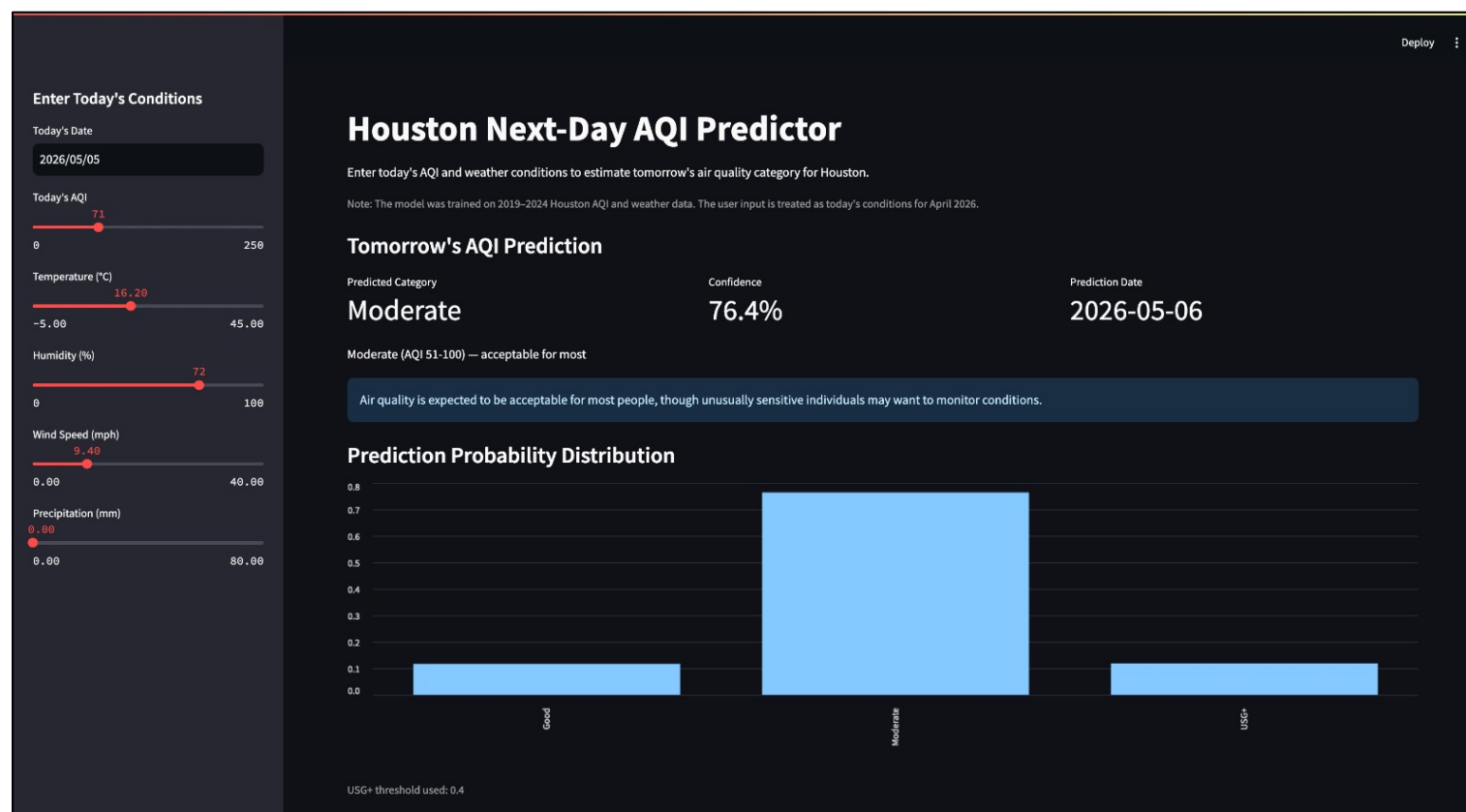
Tuned RF on TEST set:

Accuracy: 0.680

Macro F1: 0.567

	precision	recall	f1-score	support
Good	0.372	0.508	0.430	63
Moderate	0.832	0.725	0.775	327
USG+	0.424	0.596	0.496	47
accuracy			0.680	437
macro avg	0.543	0.609	0.567	437
weighted avg	0.722	0.680	0.695	437

Interactive Streamlit Dashboard



Provides an intuitive interface for users to explore air quality predictions and risk levels.



Key Considerations & Model Insights

- Real-world data is naturally imbalanced
Most days are “Moderate,” fewer harmful days
- Model focuses on detecting critical USG+ cases
Prioritizing safety over perfect accuracy
- Accuracy alone is not sufficient
Used Macro F1 and recall for better evaluation
- Balanced performance achieved through tuning
Tradeoff between overall accuracy and risk detection

Goal is to build a model that is reliable for real-world decision making

Thank you.